

Java Programming Notes with Detailed Theory and Answers

Introduction to Java

Java is a high-level, object-oriented, and platform-independent programming language developed by Sun Microsystems. It is widely used for web applications, Android applications, desktop software, cloud computing, and enterprise systems. Java follows the principle of Write Once Run Anywhere (WORA).

Questions and Answers

Q: What is Java?

Answer: Java is a programming language used for developing different types of applications. It is object-oriented, secure, robust, and platform independent.

Q: Why is Java platform independent?

Answer: Java programs are converted into bytecode, which can run on any operating system using JVM.

Features of Java

Java contains many powerful features such as simplicity, portability, security, robustness, and multithreading.

Questions and Answers

Q: What is robustness?

Answer: Robustness means the ability of Java programs to handle errors and abnormal conditions efficiently.

Q: What is multithreading?

Answer: Multithreading allows multiple tasks to execute simultaneously within a Java program.

JVM, JRE, and JDK

JVM executes Java bytecode. JRE provides runtime environment, and JDK provides development tools.

Questions and Answers

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Q: What is JDK?

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Variables and Data Types

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Q: Name primitive data types.

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Control Statements

Control statements determine the flow of execution in a Java program.

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Object-Oriented Programming

Java follows object-oriented programming concepts such as inheritance, polymorphism, abstraction, and encapsulation.

Questions and Answers

Q: What is OOP?

Answer: OOP is a programming approach based on objects and classes.

Q: What is encapsulation?

Answer: Encapsulation means binding data and methods together into a single unit.

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